

Day 3 Introduction To Ip addressing

1. List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

IPv4	IPv6
Uses 32-bit addresses.	Uses 128-bit addresses.
Example: 192.168.1.10	Example: 2001:db8::10
Provides about 4.3 billion possible addresses.	Provides a vastly larger number of addresses.
Uses decimal notation separated by dots.	Uses hexadecimal notation separated by colons.
Uses ARP to find the MAC address of a device.	Uses Neighbor Discovery Protocol (NDP) instead of ARP.
Supports broadcast communication.	Does not use broadcast ; it uses multicast and anycast instead.
NAT is commonly used because IPv4 addresses are limited.	NAT is generally not required because of the large address space.
Example in a home network: 192.168.1.1	Example: fe80::1 (link-local address).

2. Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

Static IPv4 Address Assignment

For the **192.168.10.0/24** network, I assigned unique static IPv4 addresses to all four devices. The **laptop** is assigned **192.168.10.10**, the **phone** is assigned **192.168.10.11**, the **smart TV** is assigned **192.168.10.12**, and the **printer** is assigned **192.168.10.13**. All devices use the subnet mask **255.255.255.0** and the default gateway **192.168.10.1**. This ensures that each device has a unique IP address within the same home Wi-Fi network.

3. Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.

/27 Subnetting Calculation for 192.168.1.0

Given network: **192.168.1.0/24**

New subnet mask: **/27**

1. Subnet Mask

A **/27** means 27 bits are used for the network and **5 bits** are left for hosts.

$$\textbf{/27 = 255.255.255.224}$$

The last octet is:

$$\textbf{256 - 224 = 32}$$

So, the **subnet block size = 32**.

2. Number of Valid Subnets

Starting from a **/24** network and subnetting to **/27**:

$$\textbf{Borrowed bits = 27 - 24 = 3 bits}$$

Number of subnets:

$$\textbf{2^3 = 8 subnets}$$

Therefore, there are **8 valid subnets**.

3. Hosts per Subnet

Host bits:

$$\textbf{32 - 27 = 5 bits}$$

Total addresses per subnet:

$$\textbf{2^5 = 32 addresses}$$

Two addresses are reserved:

- 1 for the **network address**
- 1 for the **broadcast address**

Usable hosts:

$$\textbf{32 - 2 = 30 usable hosts per subnet}$$

4. First Three Subnets

	Subnet Network Address	Usable Host IPs	Broadcast Address
1	192.168.1.0	192.168.1.1 – 192.168.1.30	192.168.1.31
2	192.168.1.32	192.168.1.33 – 192.168.1.62	192.168.1.63
3	192.168.1.64	192.168.1.65 – 192.168.1.94	192.168.1.95

Final Answer

CIDR: /27

Subnet Mask: 255.255.255.224

Number of Subnets: 8

Total Addresses per Subnet: 32

Usable Hosts per Subnet: 30

4. You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet.

IP Address and Subnet Calculation

Given:

- **IP Address:** 10.0.5.17
- **Subnet Mask:** 255.255.255.0

1. Convert Subnet Mask to CIDR

The subnet mask **255.255.255.0** has 24 network bits:

255.255.255.0 = /24

So, there are **8 host bits** remaining.

2. Calculate Network Address Using Bitwise AND

Convert the IP and subnet mask to binary:

IP Address: 10.0.5.17

00001010.00000000.00000101.00010001

Subnet Mask: 255.255.255.0

11111111.11111111.11111111.00000000

AND Result: 00001010.00000000.00000101.00000000

Converting the result back to decimal:

Network Address = 10.0.5.0

3. Calculate Broadcast Address

For a **/24** network, all 8 host bits are set to **1** for the broadcast address:

Network: 10.0.5.00000000

Broadcast: 10.0.5.11111111

Therefore:

Broadcast Address = 10.0.5.255

4. Valid Host Address Range

The network address and broadcast address cannot be assigned to devices.

- **First usable host:** 10.0.5.1
- **Last usable host:** 10.0.5.254

Therefore, the valid host range is:

10.0.5.1 – 10.0.5.254

5. Use an online subnet calculator tool (such as <https://www.subnet-calculator.com/>) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.

The screenshot shows the 'Subnet Calculator' interface. On the left, the 'Network Class' is set to 'A' (radio button selected). The 'IP Address' is '10.0.5.17', 'Subnet Mask' is '255.255.255.0', 'Subnet Bits' is '16', and 'Maximum Subnets' is '65536'. On the right, the 'First Octet Range' is '1 - 126', 'Hex IP Address' is '0A.00.05.11', 'Wildcard Mask' is '0.0.0.255', 'Mask Bits' is '24', and 'Hosts per Subnet' is '254'. The 'Host Address Range' is calculated as '10.0.5.1 - 10.0.5.254'. The 'Subnet ID' is '10.0.5.0' and the 'Broadcast Address' is '10.0.5.255'. The 'Subnet Bitmap' is displayed as '0nnnnnnn.ssssssss.ssssssss.hhhhhhhh'. To the right of the calculator, there is a section titled 'IP Subnet Calculator' with a description of the tool and instructions for using it.

My manual calculation **matches the output of the online subnet calculator**, confirming that the network address, broadcast address, and valid host range are correct.